

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY



Y3S2

IE3050 - Employability Skills Development

Mid-Semester Assignment

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1. Evidence-Based Self-Reflection & Lecture Impact Analysis

During my internship, one significant incident occurred while performing a vulnerability assessment on on-premises servers using Nessus and OpenVAS. I was assigned to scan a range of IP addresses and identify critical vulnerabilities. Although I successfully generated reports, I initially struggled to properly prioritize vulnerabilities based on severity and business impact. A senior engineer reviewed my work and pointed out that I had focused too much on technical severity rather than practical risk. After guidance, I revised the report by highlighting critical systems and aligning vulnerabilities with real-world impact.

This experience revealed that while I had strong technical skills, my **decision-making and prioritization skills were still developing**, which is important for employability readiness. It showed that I need to think beyond tools and focus on business context.

Self-Rating with Evidence

- **Accountability: 4/5**
I consistently completed assigned tasks such as vulnerability scans and documentation on time.
Risk: May hesitate to take full ownership in unfamiliar situations.
- **Communication: 3.5/5**
Participated in client discussions and team meetings, including explaining security findings.
Risk: May struggle to explain complex technical ideas clearly to non-technical clients.
- **Teamwork: 4/5**
Worked with senior engineers on firewall configurations and monitoring setups.
Risk: Sometimes rely too much on guidance instead of contributing ideas.
- **Adaptability: 4.5/5**
Learned multiple tools such as Graylog, ManageEngine, and Itheon PM quickly.
Risk: May take time to adapt in high-pressure situations.
- **Professional Discipline: 4/5**
Maintained documentation, followed company procedures, and met deadlines.
Risk: Could improve consistency under tight deadlines.

Lecture Impact

One lecture that had a strong impact on me was “**Application of Soft Skills**”. A key idea from this lecture was that soft skills such as communication, teamwork, adaptability, and problem-solving must be applied differently depending on the environment (academic, industrial, or multicultural).

This concept directly related to my internship experience. For example, during vulnerability assessments and client-related tasks, I realized that technical knowledge alone was not enough. I had to communicate findings clearly to team members and sometimes simplify technical details when discussing with non-technical stakeholders. I also had to adapt quickly when working with different tools, environments, and team expectations.

This lecture changed my thinking by making me understand that **being employable is not only about technical skills, but about how effectively I apply soft skills in real situations**. It helped me recognize that improving communication, teamwork, and adaptability is equally important as improving technical knowledge.

2. Industry Alignment & Professional Judgment

I reviewed a typical **entry-level Network Engineer position in a Sri Lankan IT services company** that focuses on network infrastructure, security, and system support.

Required Skills and Self-Assessment

1. Network Configuration and Troubleshooting

This includes configuring routers, switches, firewalls, and diagnosing network issues.

I meet this requirement to a good extent, as I have worked with tools and environments such as Cisco Packet Tracer, firewall configurations (FortiGate, Check Point), and WAN optimization concepts. However, I still need more real-world exposure to advanced routing protocols and live network troubleshooting.

2. Network Security and Monitoring

Employers expect knowledge in firewalls, intrusion detection, and monitoring systems.

I have strong exposure in this area through vulnerability assessments, firewall configurations, Graylog log monitoring, and Itheon Performance Monitor. I am confident in understanding threats and monitoring network activity, but I need to improve in advanced threat detection and incident response.

3. System Administration and Networking Tools

This includes working with Linux servers, Active Directory, and monitoring tools.

I have hands-on experience with Rocky Linux server setup, Active Directory configuration, and endpoint management tools like ManageEngine. While I meet the basic requirements, I need to strengthen my automation and scripting skills.

Ethical Scenario Response

If I discover a critical bug close to a deadline and my Team Lead suggests ignoring it to avoid delay, I would **not ignore the issue**. Instead, I would clearly communicate the risk and explain the potential impact of deploying a system with a critical vulnerability.

This decision reflects my values of **professional integrity, responsibility, and accountability**. Ignoring a critical issue could lead to serious security breaches and damage the company's reputation.

I believe I have the confidence to act professionally under pressure, although it may be challenging in a real-world situation. My internship experience, especially in vulnerability assessment and client-related tasks, has helped me understand the importance of making responsible decisions even under tight deadlines.

3. My Immediate Upgrade Plan

To improve my employability readiness, I plan to complete the following three specific actions within the next six months:

1. Complete Networking Certifications (CCNA / CCSA)

I will complete the CCNA certification within 4 months by following a structured study plan and practicing using Cisco Packet Tracer and real lab environments. Additionally, I will begin preparation for Check Point CCSA to strengthen my firewall and network security knowledge.

2. Develop Cloud Networking Knowledge

I will complete at least one cloud certification (such as AWS Cloud Practitioner or Azure Fundamentals) within 3 months, focusing on networking concepts such as VPCs, load balancing, and cloud security. I will also perform hands-on labs to gain practical experience.

3. Strengthen Vulnerability Assessment Skills

I will improve my skills in vulnerability assessment tools such as Nessus by completing practical labs and attempting a related certification or structured training within 2–3 months. I will document findings and practice report writing to simulate real-world scenarios.

These actions will help me strengthen both networking and cybersecurity skills required for entry-level roles.